GLOW LOGICS
SOLUTIONS

Embedded System

MICRO, SMALL & MEDIUM ENTERPRISES
सूक्ष्म, लघु एवं मध्यम उद्यम
OUR STRENGTH • हमारी शक्ति
Ministry of MSME, Govt. of India

ISO 9001 2015 CERTIFIED

TechFest
IIT BOMBAY

About Glowlogics

Glowlogics is a **government-verified MSME and Startup India-recognized company offering ISO-certified internships. In partnership with Techfest, IIT Bombay**, & **VTU BOARD**, we are committed to fostering innovation and skill development, providing practical, industry-relevant expectation

Why Choose Our Embedded Program?

- Industry-oriented curriculum designed for engineering students
- Hands-on learning with Arduino, ESP32, and sensors
- Strong focus on Embedded C programming and practical implementation
- Real-world mini project for portfolio building
- Exposure to IoT concepts and communication protocols

Program Overview

The Glowlogics Embedded Systems is a practical, industry-focused training designed for engineering students who want to build a strong foundation in embedded systems and IoT. The program covers Embedded C programming, Arduino, ESP32, sensor interfacing, communication protocols, and real-time project development through hands-on learning. By the end of the program, participants will be able to design, program, and troubleshoot embedded applications while gaining experience with industry-relevant tools and technologies.

Curriculum Overview

Module 1: Introduction to Embedded Systems

- What is an Embedded System?
- Applications in Industry
- Embedded Systems vs General Computing
- Components of an Embedded System
- Career Opportunities in Embedded & IoT

Practical

- Introduction to Wokwi & Tinkercad
- First LED Blink Simulation

Module 2: Electronics Fundamentals

- Voltage, Current, Resistance
- Ohm's Law
- Series and Parallel Circuits
- Analog vs Digital Signals
- Introduction to Electronic Components

Practical

- Virtual Circuit Building
- LED and Push Button Interface

Module 3: Embedded C Programming

- Structure of C Program
- Variables and Data Types
- Operators
- Conditional Statements
- Loops
- Arrays
- Functions
- Pointers
- Structures
- Bitwise Operations

Practical

- Logic Building Exercises
- Embedded C Programming Tasks
- Sensor Data Processing Programs

Module 4: Arduino Programming Fundamentals

- Arduino Architecture
- Digital Input & Output
- Analog Input
- PWM
- Interrupt Basics

Practical

- LED Control
- Push Button Interface
- Traffic Light Controller
- PWM Brightness Control

Module 5: Sensor & Actuator Interfacing

- Ultrasonic Sensor
- IR Sensor
- Temperature & Humidity Sensor
- LDR Sensor
- Relay Module
- Buzzer

Practical

- Distance Measurement System
- Temperature Monitoring System
- Alarm System Simulation

Module 6: Communication Protocols

- UART Communication
- I2C Communication
- SPI Communication
- Real-world Applications

Practical

- Serial Communication Simulation
- Sensor Communication Examples

Module 7: ESP32 & IoT Fundamentals

- Introduction to IoT
- ESP32 Architecture
- WiFi Communication

- Cloud Monitoring Concepts
- Smart Device Applications

Practical

- ESP32 Simulation
- WiFi Connectivity
- Sensor Data Monitoring

Module 8: Industry Mini Project

Project Options (No Hardware Required) 2 Minor , 2 Major

Beginner Projects

- Smart Traffic Light Controller
- Automatic Street Light System
- Fire Detection Alarm System
- Smart Water Tank Monitoring System

Intermediate Projects

- Smart Parking Management System
- IoT Weather Monitoring System
- Home Automation System
- Smart Classroom Automation

Advanced Showcase Projects

- Smart Agriculture Monitoring System
- Industrial Equipment Monitoring Dashboard
- Smart Energy Monitoring System
- IoT-Based Environmental Monitoring System

These are sample projects only. Unique capstone projects will be discussed in the live class

75,000+ Students

1:1 Personalized Mentorship

Taught by Industry Experts

Tools,Languages,Platforms

Wokwi

Tinkercad Circuits

Arduino IDE

Visual Studio Code

GitHub

Outcome: Students will be able to design, simulate, program, and demonstrate embedded and IoT applications independently, making them better prepared for internships, final-year projects, and entry-level embedded engineering roles.

Certificates



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CERTIFICATION OF INTERNSHIP

Congratulation, **SAMPLE NAME**

has successfully completed WEB DEVELOPMENT Internship at Glowlogics Solutions in the month of 12 MAR 2025 to 09 MAY 2025
We appreciate [his/her/their] efforts and wish [him/her/them] success in future endeavors.

Academic Head

Ministry of MSME, Govt. of India

#startupindia

ID : WD25-GL01-I009 Issue Date : 12 March 2025

SAMPLE



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Ministry of MSME, Govt. of India

CERTIFICATE

OF COURSE COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

SAMPLE NAME

has successfully completed Web Development certification with Grade A+ from Glowlogics Solutions and has proven his/her competency with utmost dedication and promise

7 April 2025
Date

Academic Head

Certificate ID: WD25-GL01-C011

SAMPLE

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Get Started Today!

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Ready to take your career to the next level?

Contact us to learn more about our courses, flexible payment plans, and how we can help you achieve your career goals.

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